Trigonometric Graphs and Models

Concept

Sine and cosine model repeating changes. Their graphs have a fixed midline and amplitude.

Model

$y = a\sin(b(x-h)) + k$, or cosine.

Features

Amplitude $|a|$; period $2\pi/|b|$; shift h; midline k.

Range

$[k-|a|, k+|a|]$; plot five points one quarter-period apart.

Tangent

Period $\pi/|b|$; unbounded range, so no amplitude.

Example 1

Describe $y = 3\sin(2(x - \pi/4)) - 1$.

Solution

1. Amplitude 3; period π; shift right $\pi/4$; midline -1.
2. Range $[-4,2]$. Quarter-period $= \pi/4$.
3. One-cycle y-values: $-1,2,-1,-4,-1$, starting at $x=\pi/4$.

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Example 2

A rider moves from 2 m to 18 m in a 40 s rotation, starting at the bottom.

Solution

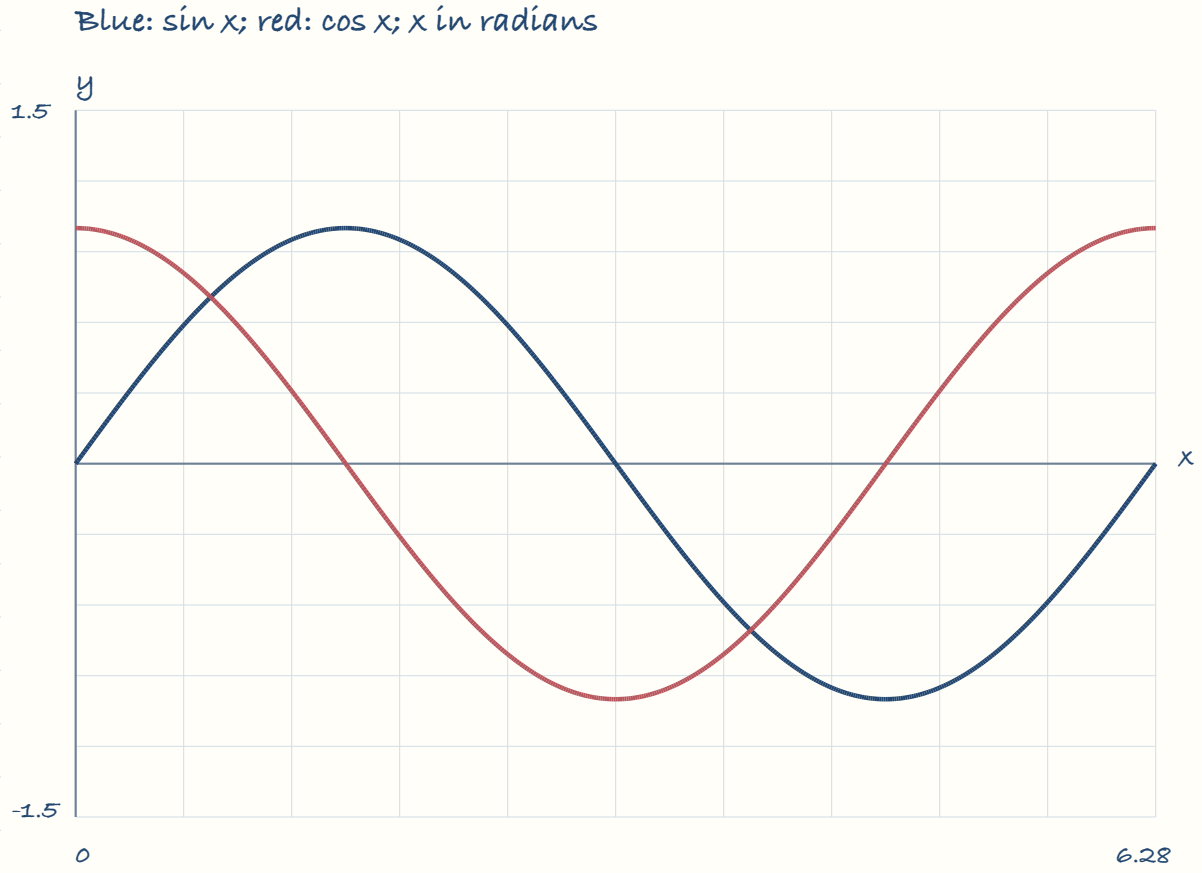
1. Midline = (18+2)/2 = 10; amplitude = (18-2)/2 = 8.

2. $b = 2\pi/40 = \pi/20$.

3. $h(t) = 10 - 8\cos(\pi t/20)$; at $t=20$, $h=18$ m.

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Visual notes | read the graph



From $x = 0$ to 2π (about 6.283), each graph completes one cycle. Both have amplitude 1 and midline 0.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. State amplitude, period and midline for $y = 4\cos(3x) + 2$.

2. Find range and period of $y = -\sin(x/2)$.

3. Write a cosine model with max 12, min 4, period 6 and a maximum at $t = 0$.

4. Explain why a sinusoid may only approximately model outdoor temperature.

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Practice answers | check after trying

1. Amplitude 4; period $2\pi/3$; midline $y = 2$.

2. Range $[-1,1]$; period 4π.

3. $y = 8 + 4\cos(\pi t/3)$.

4. Weather, seasonal shifts and unequal heating/cooling can change amplitude, timing and shape, so exact repetition is only an assumption.